

7. (2.25 points)

Given the following information:

Cumulative Reported Loss at \$1,000,000 Limit (\$000,000)

<u>Accident Year</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>
2013	450	1,354	2,465	3,348
2014	355	1,535	2,747	
2015	398	1,411		
2016	355			

Limited Expected Values at \$1,000,000 Limit (\$000)

<u>Accident Year</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>
2013	24	110	215	255
2014	25	115	214	265
2015	26	117	227	281
2016	26	121	236	290

Limited Expected Values at \$500,000 Basic Limit (\$000)

<u>Accident Year</u>	<u>12 Months</u>	<u>24 Months</u>	<u>36 Months</u>	<u>48 Months</u>
2013	24	109	195	223
2014	25	113	195	228
2015	26	115	204	239
2016	26	119	210	244

- Assume the 48 month to ultimate development factor is 1.000

a. (0.5 point)

Calculate the unadjusted 24 to ultimate volume weighted average development factor based on the reported loss triangle limited to \$1,000,000.

b. (1.75 points)

Calculate the applicable 24 to ultimate volume weighted average development factor based on the reported loss triangle limited to \$1,000,000 using the procedure outlined by Sahasrabuddhe.

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QUESTION 7			
TOTAL POINT VALUE: 2.25		LEARNING OBJECTIVE(S): A4	
SAMPLE ANSWERS			
Part a: 0.5 point			
<u>Sample 1</u> LDF(24-Ult) = (3348 / 2465) * [(2465 + 2747) / (1354 + 1535)] = 2.4503			
<u>Sample 2</u> LDF (24-36) = (2465 + 2747) / (1354 + 1535) = 1.804 LDF (36-48) = 3348 / 2465 = 1.358 LDF (24-Ult) = 1.804 * 1.358 = 2.450			
Part b: 1.75 points			
<u>Sample 1</u> Adjust triangle to basic limit			
AY	24	36	48
2013	1354 * (119/110) = 1464.78	2465 * (210/215) = 2407.67	3348 * (244/255) = 3203.58
2014	1535 * (119/115) = 1588.39	2747 * (210/214) = 2695.65	
LDF (24-36) = (2407.67 + 2695.65) / (1464.78 + 1588.39) = 1.6715 LDF (36-48) = 3203.58 / 2407.67 = 1.3306 LDF (24-Ult) = 1.6715 * 1.3306 = 2.2240			
Bring to \$1m limit = 2.2240 * [(281/244) / (117/119)] = 2.6050			
EXAMINER'S REPORT			
Candidates were expected to understand methods for estimating unpaid claims in a deductible layer, excess of a threshold, and excess of retention but bounded by a limit. Candidates were also expected to understand the interrelationships between parameters and development patterns for forecasting deductible, unlimited excess, layer excess and total claims.			
Part a			
Candidates were expected to calculate unadjusted age-to-age and age-to-ult loss development factors.			
Candidates generally did well on this question.			
Part b			
Candidates were expected to calculate the cumulative reported loss triangle at accident year 2016 cost level and basic limit. Once calculated, then the incremental and cumulative LDFs at the basic limit and accident year 2016 cost level could be computed. Finally, the volume weighted average LDFs at limit of triangle and historical cost levels could be computed.			
Common errors include:			
- Using the incorrect triangle when computing the cumulative reported loss triangle at AY 2016 cost level and basic limit. - Stopping at the cumulative volume weighted LDF at basic limit at AY 2016 cost level and			

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not continuing with the last step of calculating the volume weighted LDF at limit of triangle and historical cost level

- Using 2014 or 2016 as the 24 month LEV instead of the 2015 year.